

Akash Karthik

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SUMMARY

Embedded/controls software engineer focused on **Python** data pipelines and **C firmware** for real-time systems. I build closed-loop simulations and **software-in-the-loop** test harnesses to de-risk on-device changes, add telemetry/logging for repeatable debugging, and ship clean, testable code.

EDUCATION

San Jose State University

B.S., Computer Science

Relevant coursework: Data Structures & Algorithms; Embedded Systems; Linear Algebra; Discrete Math

San Jose, CA

Expected Dec 2026

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL

Firmware/Realtime: Fixed-point PID, anti-windup, rate limiting; CAN; deterministic timing; CSV/telemetry pipelines

Vision/Analysis: OpenCV; NumPy/pandas; latency/FPS tracking; plots/dashboards for verification

Backend: Flask/FastAPI (REST), JSON schemas, error handling; basic WebSocket/streaming

Quality/Tools: pytest unit and regression tests; scripted replays; GitHub Actions CI; Git/GitHub, VS Code, matplotlib, MoTeC i2 Pro, Busmaster, GDB

EXPERIENCE

Spartan Racing, Formula SAE

Controls & Efficiency Designer

Jul 2025 - Present

San Jose, CA

- Implemented a torque-to-power controller in C/C++ using fixed-point PID with saturation, anti-windup, and slew limits; maintained stable behavior near caps with deterministic timing.
- Built Python telemetry parsers and a **software-in-the-loop** replay harness to validate controller changes against recorded CSV logs; generated quick-look plots for parameter tuning.
- Planned and executed on-vehicle tests; profiled response and latency, documented outcomes, and iterated based on data.

Spartan Racing, Formula SAE

Power Limiting Intern

Aug 2024 - Jun 2025

San Jose, CA

- Co-developed a PID-based power limiter using torque and RPM feedback; improved endurance lap times by 5-7%.
- Extracted a desktop simulator and regression suite to reproduce field scenarios, run parameter sweeps, and prevent regressions.

Zip Intelligence

Software Development Intern — Web tooling

Sep 2024 - Oct 2024

Tracy, CA

- Implemented form validation and clearer error states; tested and debugged REST endpoints; wrote concise setup notes and shipped via PRs.

Sailfish

Data Analyst — reporting dashboards

Jan 2024 - Aug 2024

Tracy, CA

- Designed structured dashboards and weekly summaries; streamlined data entry; reported a 32% improvement in tracked metrics.

PROJECTS

EV Power Limiter

C/C++ | Firmware | Tests | CSV logging

- Fixed-point limiter with saturation, anti-windup, and slew control; Python harness **replays logs** to assert expected behavior.
- Step and ramp cases plus sweep utilities; comparison plots and a short *tuning playbook* for repeatable setup.

Vehicle Simulation

Python | Real-time sim | Telemetry/plots

- Simulator for torque, drag, acceleration, and steering; produced datasets to evaluate controller changes **offline** before hardware runs.
- Headless batch runs with seeded randomness; CSV summaries and plots for self-serve reviews.

Posture Detection System

Python | OpenCV | Real-time

- Real-time webcam pipeline with pose detection and overlays; tracked FPS/latency; smoothing/debounce and alert thresholds.
- Configurable capture pipeline (CLI flags) to enable/disable overlays, record sessions, and export labeled frames/CSV; added a small script to plot posture metrics over time for quick reviews.